



Prof. Kamakoti at DRDO Awards function

poignant memory during this period was when a distressed student approached me after failing an examination. She had come all the way from Andhra Pradesh, and her dream was shattered. I could only offer her solace by sharing that I had faced similar setbacks in my academic journey. I assured her that she could still achieve great things, which, fortunately, she found reassuring.”

In 2011, he organised India’s premier VLSI Design conference at IIT Madras, strengthening his ties with the VLSI community. Around 2012, he began pondering the question of what significant contribution he could make to his country. “I felt that we needed to move beyond academic pursuits and look for real-world, transformative projects. Dr A.P.J. Abdul Kalam’s vision for India in 2020 resonated with me. He urged us

to work on concrete solutions rather than abstract concepts, inspiring my journey towards a significant project for the country,” he explains.

At that time, the proliferation of mobile devices and surveillance cameras caught his attention. Processors were in high demand. So, he decided to explore the possibility of

creating an indigenous processor for the nation. “However, developing a processor wasn’t just about designing the hardware. We needed a robust software stack and a market for our processor. These challenges led us to explore open instruction set architectures,” he narrates.

Initially, Prof. Kamakoti and his team members considered IBM Power, which was an open instruction set architecture. However, joining the Open Power Consortium also came with financial constraints that didn’t align with their budget. Therefore, they also evaluated SPARC, but its legacy and learning curve were impractical for the academic environment. The need for a comprehensive software stack was paramount.

In 2013, they came across RISC-V, a clean-slate open instruction set architecture project led by UC Berkeley. This development sparked their interest, as it promised a serious, well-supported initiative with a complete software stack. Late one night, they decided to switch their focus to RISC-V and renamed the project Shakti, meaning ‘power’ in Sanskrit.

Between 2013 and 2017, they tirelessly worked on raising awareness and convincing funding agencies of the importance of their project. Many were unfamiliar with RISC-V, and some doubted its feasibility. How-

ever, they persevered, determined to see their vision through. During this time, they developed a basic five-stage pipeline processor as part of a student project, which gave them a solid foundation to build upon.

Rising as a thought leader

In 2017, Prof. Kamakoti’s team reached a stage where significant progress was evident. During this period, another significant development occurred with the introduction of BlueSpec, a hardware description language. BlueSpec stood out as a rule-oriented programming language designed to identify common design flaws early in the design cycle, dramatically enhancing productivity. The team started using BlueSpec in 2017 when the license cost was close to a million dollars. Fortunately, they received 250 licenses free of charge and, subsequently, BlueSpec became open source.

This shift to BlueSpec marked a pivotal moment in their Shakti project, an initiative focused on developing indigenous computer microprocessors in India. The aim of the project was to design and create microprocessors that are built in India, reducing dependence on foreign-made microprocessors, and promoting self-reliance in the field of semiconductor technology.

Subsequently, they secured funding of close to a million and a half dollars from the Ministry of Electronics & Information Technology, Government of India. That allowed them to develop two processors in phase one of the project. One of them, fabricated at 180 nanometres, was manufactured at the Semiconductor Laboratory (SCL), Chandigarh. This significant achievement showcased that India possessed the capability to produce functional chips. Their journey continued with another chip fabricated at 22 nanometres in collaboration with Intel’s Oregon fabs.

Few favourite things of Prof. Kamakoti

Music genre: Indian classical music including Carnatic music and Hindustani music

Music director: Ilayaraja

Actor: M.G. Ramachandran

Actress: Padmini

Singer: M.S. Subbulakshmi

Movie(s): Nadodi Mannan, Oppenheimer, Sholay

Comics character: Tintin

Comics book: The Castafire Emerald

Food: Lady’s finger/Okra fry